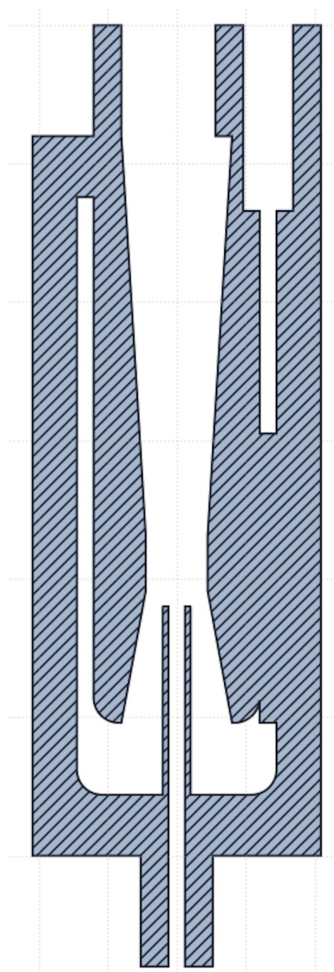
Venturi



System failures

Summary

Findings. The system forced the incoming water to reverse through almost 180° directly at the 2 mm throat, at the very point where the velocity, and therefore the dynamic pressure $\frac{1}{2}\rho v^2$, is at its maximum. Yet every local head loss is proportional to that dynamic pressure. Turning the fluid at the throat therefore greatly increases dissipation. The useful suction created by the acceleration is entirely consumed by the losses of the reversal.

Selected solution. The reversal and the acceleration are decoupled. The 180° turn takes place in an external annular space. The water is then driven into a straight internal Venturi, where it accelerates without turning. Air is admitted counter-currently through Face B by means of a coaxial needle that opens exactly at the throat.

System losses

The local head loss of a circuit fitting (bend, expansion, reversal) is expressed in terms of a loss coefficient K and of the local dynamic pressure:

$$\Delta P_s = K \cdot \frac{1}{2} \rho \cdot v^2$$

The coefficient K of a 180° reversal is of the order of 1.5 to 2.2; it depends on the geometry of the bend but not on the velocity. The whole of the dependence is carried by the term $\frac{1}{2}\rho v^2$. Performing the same geometric reversal at the throat rather than upstream multiplies the loss by the square of the velocity ratio, which is imposed by conservation of the flow rate $Q = v \cdot A$ (hence $v \propto 1/A$):

$$\frac{\Delta P_{throat}}{\Delta P_{inlet}} = \left(\frac{v_{throat}}{v_{inlet}}\right)^2 = \left(\frac{A_{inlet}}{A_{throat}}\right)^2$$

For an inlet-to-throat area ratio of between 4 and 6, the velocity ratio is 4 to 6, and the multiplying factor applied to the drag of the reversal is therefore between 16 and 36. At the throat, the system dissipates 16 to 36 times more energy than the same turn would have cost had it been placed upstream.

Comparative physical analysis

Extended Bernoulli equation (inlet → throat → outlet)

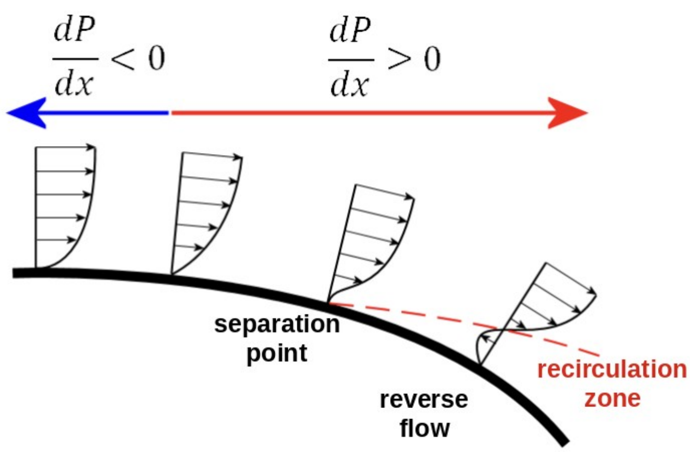
The suction available at the throat to draw in air is the difference in static pressure between the upstream section and the throat.

$$\Delta P_{throat} = P_1 - P_2 = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right) - \Delta P_{losses}^{1\to 2}$$

The term $\frac{1}{2}\rho(v_2^2 - v_1^2)$ is identical for both systems (same throat geometry, same flow rate). The entire difference is played out in ΔP_{losses}. In System 1 this term is dominated by the loss of the reversal, evaluated at v_{throat}; it may become as large as the driving term, which can cancel ΔP_{throat} altogether.

Boundary layer separation and the ceiling on suction

Beyond the throat, the fluid must decelerate in order to convert kinetic energy back into static pressure (recovery). This deceleration imposes an adverse pressure gradient ($dp/dx > 0$). Close to the wall, within the viscous sub-layer, the fluid possesses very little momentum; it is the first to be brought to rest and then reversed by the adverse gradient. When the wall velocity profile vanishes ($\partial u/\partial y|_{wall} = 0$), the boundary layer separates: a reverse flow and a recirculation zone are formed.



System 1. The reversal at the throat superimposes a turn of almost 180°, of pronounced relative curvature, upon the adverse gradient of the recovery. The adverse gradient becomes strong and highly localised. Separation is significant. The actual suction at the throat plateaus below its ideal value: there is a saturation effect that is independent of the driving power. Increasing the flow rate no longer increases the suction; it merely feeds recirculation and noise.

System 2. The straight diffuser keeps dp/dx gentle enough for the boundary layer to remain attached over its entire length. The recovery of static pressure proceeds monotonically and the recovery coefficient C_p approaches its ideal limit:

$$C_p = \frac{P_{outlet} - P_{throat}}{\frac{1}{2}\rho v_{throat}^2} \quad , \qquad\qquad C_{p,\ ideal} = 1 - \left(\frac{A_{throat}}{A_{outlet}}\right)^2$$

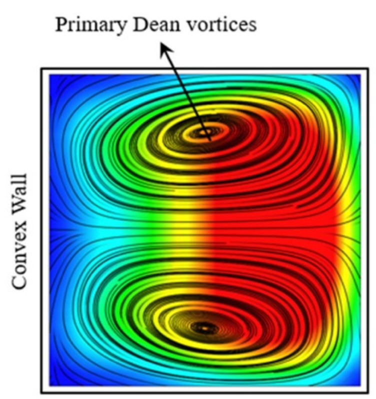
Hydrodynamic instabilities: Dean secondary flows

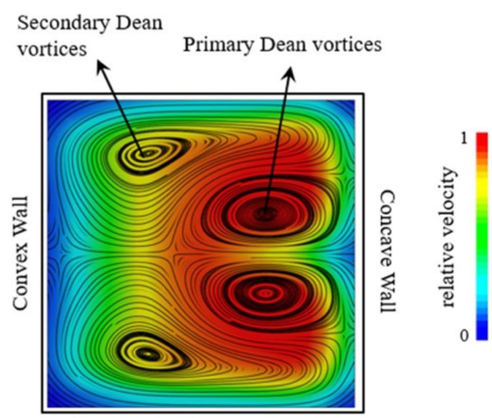
In any curved duct, the balance between the centrifugal force and the radial pressure gradient cannot be achieved simultaneously for the fast-moving core and for the slow fluid at the wall. The result is a pair of counter-rotating vortices (Dean vortices) superimposed upon the main flow. Their intensity is governed by the Dean number:

$$De = Re\sqrt{\frac{D_h}{2\,R_c}} \quad ,$$

System 1. The reversal takes place at the throat, at high Re and with a very small radius of curvature R_c: the Dean number is considerable and the secondary vortices are intense, asymmetric and unsteady. They distort the profile at the throat, cause the point of air injection to fluctuate and render the suction erratic.

System 2. The reversal takes place within the annulus at low velocity, which sharply reduces De. Three guide vanes set at 120° partition the annulus and break up the azimuthal behaviour of the Dean cells: they prevent the formation of extended vortices and stabilise the flow around the central Venturi. The flow through the reversal becomes regular and axisymmetric once more, a necessary condition for stable air injection.





Theoretical justification of the sizing of System 2

The ‘reversal / acceleration’ decoupling principle

This consists in separating in space the two operations that System 1 superimposed at the throat. Since the loss of the reversal is proportional to v^2, it is imposed on the fluid where the fluid is slowest (the annulus), the fluid then being accelerated where it is no longer turning (the straight Venturi). The sizing of the annulus follows from this:

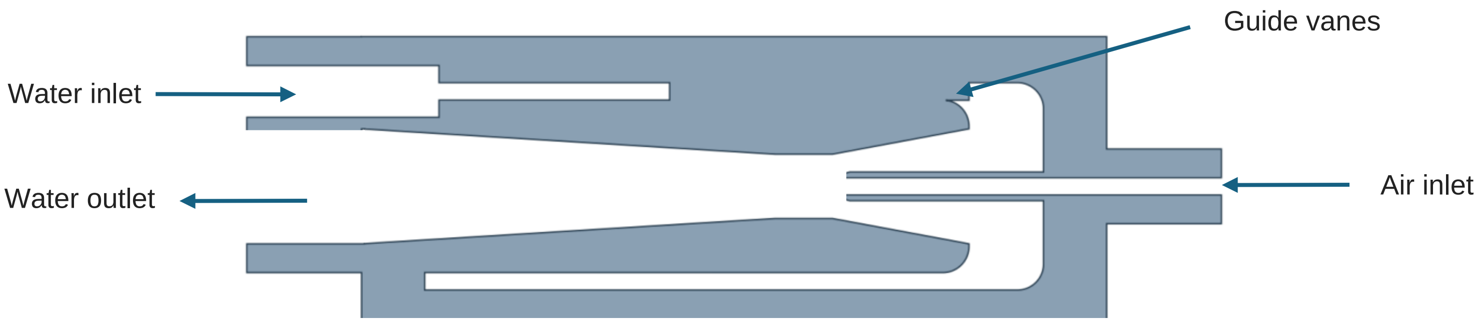
$$4\,A_{throat} \leq A_{annular} \Rightarrow v_{ann} = \frac{Q}{A_{ann}} \leq \frac{1}{4} v_{throat} \Rightarrow \frac{1}{2}\rho\, v_{ann}^2 \leq \frac{1}{16} \cdot \frac{1}{2}\rho\, v_{throat}^2$$

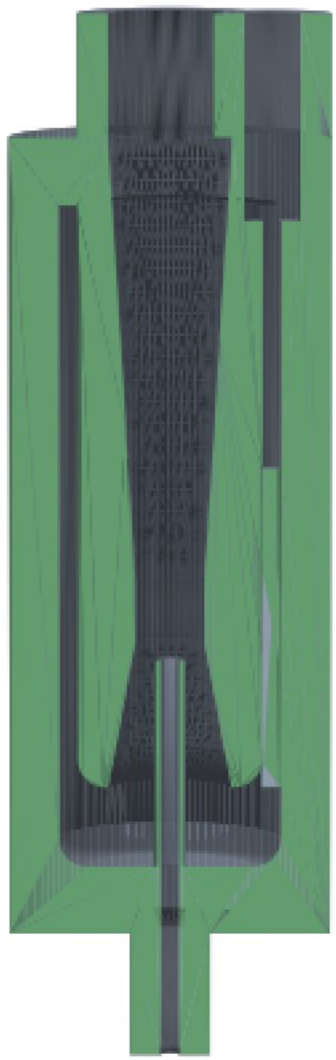
The annular area is set at four times the throat area: the velocity there is therefore less than a quarter of v_throat, and the dynamic pressure, and hence the loss of the reversal, is reduced by a factor of 16.

Coaxial alignment of the air needle

The needle is a slender component built into the base of Face B and opening at the throat. Any departure from coaxiality makes the annular water passage at the throat asymmetric: the velocity becomes higher on the side of reduced clearance, creating a lateral pressure imbalance, premature one-sided cavitation and deflection of the jet.

Appendix A — Model





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